

3.4.3 Bitwise Operators

بٹ وائز آپریٹرز (Bitwise Operators) اعداد کو بائنری میں بدل کر ہر بٹ پر الگ الگ کام کرتے ہیں۔

CS XI

Chapter 3: Programming Fundamentals

Lecture 12 of 24

★ Golden

3.4.3 Bitwise Operators ★ Golden

LEARN IT

6. Bitwise operators treat numbers as **binary digits** (bits: 0s and 1s) and work on them **bit by bit**, column by column. Think of each bit as a switch that is ON (1) or OFF (0). Python's `bin()` function shows the binary form of a number (it starts with `0b`).

- **AND (&)** — gives 1 **only if both** bits are 1, otherwise 0.
- **OR (|)** — gives 1 if **at least one** bit is 1.
- **XOR (^)** — gives 1 if the bits are **different**, 0 if they are the **same**.
- **NOT (~)** — inverts all bits (0 → 1, 1 → 0). In Python the result is the number made negative, minus 1: $\sim x = -(x + 1)$. So ~ 10 gives -11 .
- **Left shift (<<)** — moves bits left and adds 0s on the right. Each shift **multiplies by 2**.
- **Right shift (>>)** — moves bits right and removes the rightmost bits. Each shift **divides by 2** (floor division).

Bit A	Bit B	A & B	A B	A ^ B	~A
0	0	0	0	0	1
0	1	0	1	1	1
1	0	0	1	1	0
1	1	1	1	0	0

WORKED EXAMPLE

Worked example: a = 10, b = 6. First convert to binary using place values 8, 4, 2, 1: $10 = 8 + 2 = 1010$; $6 = 4 + 2 = 0110$. Then work column by column.

	8	4	2	1	Decimal
a = 10	1	0	1	0	10
b = 6	0	1	1	0	6
a & b	0	0	1	0	2
a b	1	1	1	0	14
a ^ b	1	1	0	0	12

- AND: only the "2" column has 1 in both → 0010 = **2**.
- OR: columns 8, 4 and 2 have at least one 1 → 1110 = 8 + 4 + 2 = **14**.
- XOR: columns 8 and 4 are different; column 2 is the same (both 1) → 1100 = 8 + 4 = **12**.
- NOT: $\sim 10 = -(10 + 1) = \mathbf{-11}$.

```
a = 10
b = 6
print(bin(a), bin(b))
print(a & b)
print(a | b)
print(a ^ b)
print(~a)
```

OUTPUT

```
0b1010 0b110
2
14
12
-11
```

Worked example: shifts. 5 in 8 bits is **0000 0101**.

Expression	Binary	Decimal	What happened
5	0000 0101	5	start
5 << 1	0000 1010	10	add one 0 on the right (5 × 2)
5 << 2	0001 0100	20	add 00 on the right (5 × 4)
20 >> 1	0000 1010	10	remove the rightmost bit (20 ÷ 2)
20 >> 2	0000 0101	5	remove two bits (20 ÷ 4)

```
a = 5          # Binary: 0000 0101
print(a << 1)   # 0000 1010
print(a << 2)   # 0001 0100
b = 20         # Binary: 0001 0100
print(b >> 1)   # 0000 1010
print(b >> 2)   # 0000 0101
print(bin(20 >> 2))
```

OUTPUT

```
10
20
10
5
0b101
```

REMEMBER IT

Shortcuts: $x \ll n = x \times 2^n$, $x \gg n = x // 2^n$, $\sim x = -x - 1$. For AND/OR/XOR always write both numbers in binary, one under the other, with the same number of bits.

COMMON MISTAKE

Do not mix up the logical and bitwise operators: `and` / `or` / `not` work on True/False conditions, while `&` / `|` / `~` work on the bits of integers. Also remember `bin(6)` prints `0b110`, without the leading zero — add zeros yourself when lining up columns.

اردو میں سمجھیں

بٹ وائز آپریٹرز (Bitwise Operators) اعداد کو بائنری میں بدل کر بٹ پر الگ الگ کام کرتے ہیں۔ `&` تب 1 دیتا ہے جب دونوں بٹ 1 ہوں، `|` تب جب کوئی ایک بھی 1 ہو، اور `^` تب جب دونوں بٹ مختلف ہوں۔ `~` تمام بٹ الٹ دیتا ہے، اس لیے `~10` کا جواب `-11` ہے۔ لیفٹ شفٹ عدد کو 2 سے ضرب دیتا ہے اور رائٹ شفٹ 2 سے تقسیم کرتا ہے۔

EXAM TIP

Exams ask you to "evaluate `12 & 10`, `12 | 10`, `12 ^ 10`, `~5`, `7 >> 1` showing binary working". Marks are given for the binary conversion and column working, not just the final number. Always draw the column table.

CHECK YOURSELF

1. Predict the output (show the binary working):

```
print(12 & 10, 12 | 10, 12 ^ 10)
print(~5, 7 >> 1, 3 << 3)
```

2. What does `bin(13)` print?

Answers: 1. `8 14 6` and `-6 3 24`. `12 = 1100`, `10 = 1010`: AND = `1000` = 8, OR = `1110` = 14, XOR = `0110` = 6. `~5 = -(5 + 1) = -6`. `7 = 111 → 11 = 3`. `3 = 11 → 11000 = 24 (3 × 8)`. 2. `0b1101 (8 + 4 + 1)`.

Next lecture: 3.5 Control Structures